

Neural Network-Based Predictions of Supersymmetric Contributions to the Anomalous Magnetic Moment of the Muon (Abstract)

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In this work, we analyze the well-known anomaly in the muon's magnetic moment using a flavor-extended Minimal Supersymmetric Standard Model (MSSM) with a hierarchical family structure in the trilinear scalar soft-supersymmetric terms of the Lagrangian, defined at the SUSY breaking scale. This framework introduces five free coupling parameters within specific ranges, with the goal of reducing the discrepancy between theory and experiment to the range $[3.15\sigma, 4.15\sigma]$.

Using Neural Networks (NN), we predict the supersymmetric contribution a_μ^{SUSY} by generating truncated Gaussian distributions for the five SUSY parameters, ensuring values remained within physically motivated parameter ranges and selecting those within the desired deviation range. The selected parameter sets serve as input for the neural network, while the corresponding values a_μ^{SUSY} , explicitly computed from the SUSY model, are used as target outputs for training.

The trained model successfully achieved the primary objective, yielding acceptable predictions with low MSE and MAE values. The analysis revealed a strong relationship between certain coupling constants and their corresponding probability distributions. This approach provides a fast alternative to traditional analytical computations for obtaining phenomenological predictions, though further improvements such as Bayesian Neural Networks, increased layers, and Monte Carlo simulations are suggested as future work.